

ProofRoad

A truck drives only as fast as the road ahead has proved itself clear.

Smart India Hackathon 2026 | Problem statement 26007 | Safe and efficient operation of mine vehicles in fog | Ministry of Steel / NMDC Limited | Version 2

The idea in one breath

In fog, a truck should only go as fast as the distance it can prove is clear allows. Every piece of information (own radar, another truck's message, a fog sensor, a road gate) counts as proof only while it is fresh. Old or missing information never counts as "clear".

Analogy. Indian Railways lets one train into a track section at a time and treats "no signal" as "stop". ProofRoad does the same for mine haul roads, without a control room in the safety loop.

The problem

At Bailadila in monsoon, visibility on hilltop haul roads can fall to 3 to 5 m, while a loaded dumper can need many times that to stop. Today's two answers are both bad: run blind (safe only by luck) or stop everything (ore stops moving and shifts are lost). ProofRoad measures how much road is **proven** clear, sets the speed to match, and slows only the stretch that is actually risky.

The rule that keeps it safe

Fresh evidence can shorten the proven distance at once. Only fresh proof can lengthen it, and **unknown is never counted as clear**. If the network, camera, fog post or dashboard dies, the truck's own radar warning still works. Other parts can make a truck more careful, never less careful than its own radar says.

The pitch in one line

"We did not add another dashboard. We built a system that can explain exactly why the next metre is safe, and becomes conservative when it cannot."

What is new compared with the source designs

- Message freshness by age counters, so no shared clock is needed.
- Hazards leave the list only when a truck actually looks, never by a timer.
- Right-of-way comes from how much room each truck has left to stop, not from a fixed loaded-wins rule.
- Fog posts are audited by the cameras of trucks that drive past.
- A virtual-fleet simulation gives efficiency numbers that two model trucks cannot.
- **[NEW]** Phantom Obstacle view: a hazard the driver cannot see is drawn on the road strip with its distance.
- **[NEW]** A dedicated safety chip is a stated design rule: the radar warning never depends on the AI computer.

What changed in v2

Version 2 folds in the few ideas from the AegisMesh and FogNet Sentinel documents that fit the fixed must-build list. Everything else in those documents was reviewed and deliberately left out (see the source page). Changed items are shaded.

Change	From	Where it lands	Extra build effort
Dedicated safety MCU owns the radar loop, corridor math, watchdog and buzzer. Pi/laptop does camera, AI and dashboard only.	AegisMesh (Tier 1)	Truck brain, architecture page	Near zero (already planned)
Phantom Obstacle: a follower sees the hazard as a marker with distance on its road strip.	AegisMesh (SVPP)	Hazard list, driver screen, demo 3:00	Small (UI on existing data)
Fixed driver screen: state safe speed clear distance needed distance.	FogNet Sentinel (HMI)	Driver screen	Small
Near-miss log added to the replay black box.	Sentinel / FogNet 4.1	Advise and log, dashboard	Small
Emergency beacon carried by a supervisor, signed, bypasses the control room. Still optional.	AegisMesh	Build list, judge Q&A	Optional
Idle trucks as temporary fog observers: simulation-only, not a demo claim.	AegisMesh	Flaws table	None (not built)
Worst-case table for 3 to 5 m fog, and why radar-first beats thermal/LiDAR here.	Review of the problem statement	Decision page, flaws	None (paper)

Scope guard. No new hardware was added. Every change above is either already in the plan, a screen change, or a paper change. The must-build list and the cut list on the demo page are unchanged.

Four parts, one rule

Layer	What is in it
Ops dashboard (watches and replays only)	Closures, analytics, event replay, near-miss trends
Fog posts	Visibility, trend, own health
Blind-curve gates	One direction at a time
Shared mesh	Truck beacons, hazard list, gate tokens, signed emergency stop
Truck safety loop (works with no network)	Sense: radar first, IMU, camera SafeCorridor: proven vs needed distance Advise and log: speed, reason, replay

The truck safety loop owns the safety decision. Everything else adds evidence and can only add caution.

Part	What it does
Truck brain (two chips)	[NEW] Safety MCU (STM32/ESP32): reads radar, runs the SafeCorridor formula, drives the buzzer and vibration, and has a hardware watchdog. It has no dependency on anything else. Assist computer (Pi or laptop): camera, object naming, fog check, dashboard link. If the assist computer or camera dies, the MCU's radar warning and buzzer keep working. The MCU alone can show a minimal speed and warning on a small display.
Fog posts	Roadside boxes measure how far light can travel, plus temperature and humidity, and report whether fog is rising, steady or falling. We report current visibility and trend, not a 20-minute forecast.
Shared hazard list and messages	Trucks share small position-tagged messages carrying their age and a signature. Old or forged ones are dropped. Followers get early warning of obstacles they cannot see yet, shown as a Phantom Obstacle marker.
Blind-curve gates	Two gates at a single-lane blind stretch let one direction in at a time. If they are unsure, they say wait.
Emergency beacon (optional)	[NEW] Carried by a blast supervisor or shift lead. One button sends a signed stop message that every truck shows as a critical alert, without going through a control room. Unsigned stop messages are rejected.

Messages travel over ESP-NOW (fast beacons, about 10 per second). LoRa carries only slow traffic: fog posts, the emergency beacon and long-range relay. The truck's radar loop depends on neither.

Why radar-first (and not thermal or LiDAR) at 3 to 5 m visibility

[NEW] Thermal cameras and LiDAR are on the problem statement's list, but both lose range in dense fog and heavy rain droplets, and LiDAR is the most affected. Millimetre-wave radar is the least affected by fog, so it is the primary sensor. A thermal camera is a sensible optional add-on for naming warm objects (people, engines) once radar has found something, but no safety decision depends on it. Range numbers in this document are targets until bench-tested.

How one decision is made

1	Measure.	Radar tracks obstacles ahead and places them on the known route, so a truck on another branch is not mistaken for one on this road.
2	Prove.	Proven-clear distance is the shortest of three: the radar's healthy range, the nearest hazard or peer truck, and the first stretch nobody has proved (a blind curve without a gate OK, a map doubt, growing position error).
3	Need.	Needed distance adds driver reaction, braking and a buffer. Braking comes from tests on this class of truck at this load, slope and surface, always using the pessimistic value.
4	Compare and advise.	If proven is at least needed, the truck holds speed. If not, the screen shows a lower speed worked out from the same formula.
5	Show and log.	One speed, one arrow, one reason. Ten seconds before and twenty seconds after every warning are saved so a supervisor can replay exactly why. Near misses are also logged.

Distance needed = speed x reaction time + speed² / (2 x braking) + buffer.

Braking = tested brake value - slope effect - wet-road penalty.

Safe speed = braking x [-reaction + sqrt(reaction² + 2 x (proven - buffer) / braking)]. If proven is not more than the buffer, safe speed is zero: stop.

Worked example (illustrative numbers)

A loaded truck on a wet downhill drives at 30 km/h. Reaction time 1.5 s, buffer 10 m, braking 1.4 m/s² (2.5 minus 0.3 wet minus 0.78 slope).

Piece	Distance
Driver reaction (8.3 m/s x 1.5 s)	12.5 m
Braking (8.3 ² / 2.8)	25 m
Safety buffer	10 m
Truck needs about	47 m
Radar proves clear	30 m

47 m needed is more than 30 m proven, so the screen shows: "Slow to 20 km/h. Reason: only 30 m proven clear." With 40 m usable, an empty truck on a dry flat road (braking 3.0) may go about 42 km/h, while the loaded wet downhill truck may go about 31 km/h. Same radar range, different safe speed. A fixed "warn at 20 m" rule cannot do this.

[NEW] Worst case: what 3 to 5 m fog really means

Driver visibility of 3 to 5 m is not the same as proven distance. The driver cannot see 30 m, but healthy radar can prove it. The dangerous case is when proven distance itself collapses: a blind curve with no gate OK, a stale hazard, degraded radar, or heavy rain clutter. Same formulas, buffer 10 m, reaction 1.5 s:

Proven clear	Loaded, wet, downhill (braking 1.4)	Empty, dry, flat (braking 3.0)	Screen shows
100 m	50 km/h (site cap applies)	69 km/h (site cap applies)	Site speed cap
60 m	36 km/h	48 km/h	Advised speed
30 m	20 km/h	26 km/h	Advised speed
15 m	8 km/h	9 km/h	Creep, warn
10 m or less	0 (stop)	0 (stop)	STOP: not enough proven road

Reading the table honestly: when proven distance falls to the buffer, the truck stops and waits for proof (a passing truck, a gate OK, a supervisor). It does not creep blind. The productivity gain comes from the many stretches where radar proves 30 m or more; the safety gain comes from refusing to move when it does not. The site sets the buffer, and a smaller buffer at crawl speed is a site decision, not ours.

Blind curve: one direction at a time

Each gate asks each truck one question: how much room do you still have before the point where you can no longer stop? The truck with the least room goes first, because it cannot afford to yield. The other holds only if it can still stop.

Case	Room left A / B	Result
Loaded meets empty	6 m / 22 m	A goes, B holds
Empty truck already committed	18 m / -1 m	B goes, A holds
Gate message lost	any	Both wait. Uncertain is never clear. A truck already inside exits slowly on its own radar
Neither can stop in time	-4 m / -2 m	Both stop now. Logged as a design-level failure

Gate states: clear, reserved, occupied, uncertain, closed. A gate only clears after two independent signs the truck has left (gate sensor and the truck's own message).

Shared hazard list: cleared by looking, not by a timer

Step	What happens	Following truck sees
Spotted	Truck A's radar tracks something for several frames. It joins the list with a road position, unverified.	Possible hazard, approach with care
Confirmed	A second independent source (another truck's radar) sees it.	[NEW] Phantom Obstacle marker: hazard confirmed 120 m ahead, advised 12 km/h
Nobody looks	No truck has passed for a while. The record becomes stale, not clear.	Old report, approach slowly until someone looks
A truck passes	A healthy-radar truck drives through and sees nothing along the whole pass. Confirmed hazards need a second pass or a supervisor OK.	Being re-checked
Cleared	Removed because trucks actually looked.	Road proven clear, normal speed

Why not a timer? A timer would turn the hazard green after a few seconds of silence even if the rock is still there. Here silence only means "old report", and the next truck's radar heals it.

[NEW] Driver screen layout

The screen never shows a confidence percentage. It shows one state, one speed, and the two distances that caused it, so the driver sees the reason at a glance.

SLOW DOWN	SAFE SPEED: 8 km/h	CLEAR 18 m NEED 25 m
Road strip: [YOUR TRUCK] ---- 18 m ---- [HAZARD marker] ---- (unproven)		

Example values only. Escalation is immediate; returning to normal needs stable health, a waiting period and driver acknowledgement.

Fog posts get checked by the trucks that pass them

A dusty or drifting fog post could say "clear" while the road is white. So every passing truck estimates visibility itself and compares.

Topic	How it works
Camera fade test	Two high-contrast boards stand at known distances. The camera reads how faded each looks. Comparing the two cancels the unknown brightness and gives visibility of roughly 3.9 divided by the fading rate per metre (Koschmieder relation, 2% contrast threshold). Experimental, needs calibration.
Who wins a disagreement	Post says clear but trucks see dense fog: trucks win at once and the post is flagged for cleaning. Post says foggy but trucks see clear: the road stays cautious until several trucks agree. Mistakes fall on the careful side.
Fog or dust	Fog needs a small gap between air temperature and dew point. Bright scatter with dry air is tagged dust. Both cut what the driver sees, so the speed cap does not depend on the label.
Trustworthy messages	Each message carries age, sequence number and a signature. Age counters replace a shared clock, which a GPS-poor pit does not have. Old copies are rejected and logged. Messages can only add caution, so a forged one causes a nuisance slowdown, never a hidden hazard.

When things break, the truck gets more careful

If this fails	The truck does
Network	Keeps radar and its own speed advice. Peer information goes stale and stops counting. Shows "cooperative link lost".
Camera or assist computer	[NEW] Safety MCU and radar continue unchanged. Health warning shown.
Fog post	Uses radar and the last known cap. A missing post never raises the speed.
Gate or its message	Gate says wait and trucks slow down.
Radar blocked	Shows "degraded, stop at a safe place". No other sensor pretends to replace it.
Fake or old message	Rejected and logged. Local warnings unaffected.

Warnings escalate at once. Returning to normal needs stable health, a waiting period and driver acknowledgement, so one good frame cannot cancel a real problem.

The six-minute demo

Time	What judges see	Proves
0:00	Two model trucks move. Screen green: "60 m proven clear, safe up to site cap".	Baseline
0:40	Mist starts. Fog post flags rising fog and advised speed drops before the truck arrives.	Early warning
1:30	Stopped obstacle hides in mist. Camera cannot see it, radar can, and advice falls with the reason shown.	Radar-first proof
2:15	Switch to loaded and wet. Same distance, lower safe speed.	Braking model
3:00	Truck A spots a hazard. Truck B behind sees [NEW] a Phantom Obstacle marker, "hazard 80 m ahead". A truck passes, sees it gone, the list clears.	Shared hazard list
3:45	Two trucks meet at the blind curve. One GO, one HOLD.	Gates
4:30	Network off, then camera covered. [NEW] The safety MCU keeps the radar warning and buzzer running while the assist computer is dead.	Nothing fails silently
5:15	Replay a fake old message (rejected). Block the radar: truck shows "degraded".	Honest failure
5:45	Replay screen shows why each decision was made, including a near-miss entry. Then the fleet simulation against "slow to a crawl".	Explainability, efficiency

The fleet simulation adds 10 to 20 virtual trucks on the laptop using the same messages. It is labelled a simulation and reports throughput against a "crawl in fog" policy using stated assumptions.

What we build and what it costs

Item	Qty	Note
Model trucks: radar or labelled stand-in, IMU, wheel encoder, radios, screen, buzzer, safety MCU	2	Braking measured by stopping carts with weights
Fog posts: scatter sensor, temperature and humidity, LoRa	2	Battery or solar
Blind-curve gates: lamp and break-beam sensor	2	No central broker
Emergency beacon (optional)	1	Signed stop message, critical on every screen
Laptop dashboard, mist source, track, curve, obstacle, fog-check boards	1 set	Replay and analytics only
Planning total		About Rs. 40,000 to 60,000, or about Rs. 60,000 to 85,000 with one 60 GHz evaluation radar. Estimates: verify prices and lead times.

Must build - Truck brain: safety MCU with measured braking - Radar, screen, buzzer - Message age and replay rejection - Fog post with trend - Network, camera, radar failure demos	Then, in this order - Shared hazard list with Phantom Obstacle marker - Blind-curve gates - Fleet simulation - Camera fog check - Near-miss log entries in replay	Cut - 3D dashboard - Safety score - Worker tags - Escort mode - RadarGlyph reflectors - Idle-truck observer nodes (simulation only) - Patent wording and any "100%" claim
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Freeze the scenario list first. After that, add a feature only to fix a failure you measured.

Flaws we have, said before you ask

"Matters" is our own judgement of how much each could hurt if ignored.

Flaw	What we do about it	Matters
Demo radar is a stand-in	It proves the logic, not mine-grade range. Mines need certified 77 GHz radar. We bench-test the stand-in on a stopped metal target from a moving cart and label it clearly.	High
Braking numbers are ours	They come from stopping our carts. A real mine must test its own trucks with the maker. We use the pessimistic value, and evidence can only lower the allowed speed automatically.	High
Radio in a pit is unproven	Benches shadow and reflect signals. We run range tests, let any truck relay once, and go careful when links degrade. Local radar never depends on it.	High
Gates could jam or false-clear	Uncertain blocks entry, exit needs two independent signs, and a manual clear procedure exists. Loss and delay cases are in the test matrix.	High
Trusting a leader's earlier pass	Letting followers go faster because a truck just passed could miss something that appeared afterwards. Off by default and simulation-only until field-tested.	High
Feature creep	The must-build list is fixed and nothing is added after the freeze except fixes for measured failures. v2 added no hardware.	High
[NEW] Fog this dense may leave little proven road	At 3 to 5 m visibility the truck depends entirely on radar range. If proven distance drops to the buffer the truck stops. That is correct behaviour but it costs cycle time; the fleet simulation shows how much.	High
[NEW] Idle trucks as fog observers	A parked truck is often not on the fog-prone haul road and its sensors would need calibration. Kept as future work and simulation-only. We do not claim it in the demo.	Medium
Mist is not natural fog	Fog-machine droplets differ, so the sensor curve needs re-tuning on site. We claim repeatability, not mine equivalence.	Medium
Only two physical trucks	Fleet behavior comes from a labelled simulation on the same messages, not real fleet statistics.	Medium
No fog forecast	We measure current visibility and trend. Forecast probabilities need seasons of site data.	Medium
Prototype security is basic	Shared keys are fine for a demo. Production needs per-device keys and secure chips. The emergency beacon uses the same signing.	Medium
We advise, we do not brake	The demo relay is a labelled mock. Real intervention needs the truck maker's approval and functional-safety work (ISO 21815, ISO 19014). By design.	By design

Judge questions, answered plainly

Question	Answer
What is actually new here?	Every source of information has an age and a proof value. Only fresh proof lets a truck go faster, unknown never counts as clear, and hazards are removed by someone looking, not by a timer. Radar, V2V and fog sensors are not new alone; the combination and the rules are.
What if LoRa fails?	LoRa is not the safety link. Fast messages use ESP-NOW, and the truck's radar loop is independent of both. Lost links only make the truck more careful.
Why the loaded truck goes first?	It is harder to stop, and we work that out from distances instead of a fixed rule. If an empty truck has already committed, it goes first. A site rule can override.
Why not AI cameras?	Cameras degrade in fog. Radar decides, and AI only helps name what the radar found. The safety state is a formula you can replay.
Why not thermal or LiDAR?	[NEW] Both lose range in dense fog and rain, LiDAR most. Radar is least affected, so it is primary. Thermal is an optional add-on for naming warm objects, never a safety dependency.
What if the AI computer crashes?	[NEW] A separate safety MCU runs the radar loop, the formula, the watchdog and the buzzer. It does not need the AI computer.
What if GPS is wrong?	We track position along a known road with checkpoints, so GPS is optional. Growing position error shortens the proven distance and lowers the advised speed.
How does a driver see a hazard he cannot?	[NEW] The follower's screen draws a Phantom Obstacle marker at the hazard's road position with its distance, fed by a small position-tagged message, not video.
Who can stop all trucks?	[NEW] Only a signed emergency beacon carried by an authorised person. Unsigned stop messages are rejected.
Can it brake a real truck?	No. It is advisory. Real intervention needs the truck maker's interface, functional-safety analysis and mine approval.
Is it AI?	AI names objects and can help with fog trends later. The decision to slow down comes from the stopping formula.
How does it help productivity?	Clear stretches keep full speed and only risky segments are slowed. The fleet simulation reports the difference against slowing everything, using stated assumptions.

Where every idea came from

ProofRoad is a synthesis of eight documents: six original solution designs plus two later syntheses (AegisMesh and FogNet Sentinel). A seventh original design was mentioned but not received. Titles and names are as they appear in the submitted file names. **Check your event's attribution rules before submitting this page.**

Source document	What we kept	What we dropped
FogNet Nexus (Chatgpt_FogNet_Nexus.pdf, drafted with ChatGPT)	Green/yellow/red communication health. Movement-token idea. Stale-message rejection. Safety black box.	LoRa-only vehicle messaging. Alpha escort mode.
FogNet Alive (Claude_FogNet_Alive.pdf, drafted with Claude)	Passing trucks auditing fog posts. "A role, not an identity." Adaptive sampling. Emergency beacon. Stating limits before being asked.	Ultrasonic as main sensor. Signal-strength-only distance. Loaded truck always wins.
FOGNET 4.1 (AboliFogNet_4_1.pdf, by Aboli)	Explainable alerts. Safety states. Near-miss logging. "AI predicts, physics decides."	Worker tags and hazard beacons in the core.
FOG-SHIELD X (PrathameshFOG_SHIELD_X.pdf, by Prathamesh)	Shared hazard record with segment, severity, confirmations. Early warning for followers.	Invented percentage trust scores. Timer-based clearing. Mine Safety Index.
NEXORA FogNet SafeEnvelope (Snehal_NEXORA_FogNe.pdf, by Snehal)	Proven-clear corridor against stopping need. Measured braking bins. Fail-closed blind-curve block. Signed, age-checked messages. Test matrix. Honest claims.	Radar-reflector pattern decoding. Central broker. The larger budget and scope.
FogNet Swarm (GLM_Fogswam.docx, drafted with GLM)	Small position-tagged hazard packets shown on a following truck's screen. Mass-score right-of-way idea (replaced by room-to-stop).	Two-wavelength "100%" fog/dust claim. 3D fog cloud. Unauthenticated "Stop All".
[NEW] AegisMesh: Cooperative Kinetic Safety Architecture	Dedicated safety MCU tier. Phantom Obstacle presentation of shared hazards. Supervisor-carried emergency beacon. Idle-truck observer (as simulation-only future work).	RadarGlyph reflectors. Kinetic Priority score (units do not balance). IR/red fog-vs-dust claim. Worker tags. 77 GHz + RTK-GNSS cost base.
[NEW] FogNet Sentinel (FogNet_Sentinel_Final_Solution.pdf)	Driver screen layout with clear and needed distance. Safety Black Box with near-miss log. Local-safety-first hierarchy.	Nothing further; it is a subset of ideas already here.

New in ProofRoad: message freshness by age counters instead of a shared clock; hazards cleared by a truck passing; right-of-way from remaining room to stop; fog posts audited by the camera fade test; a virtual-fleet simulation for efficiency numbers; trusting a leader's earlier pass, held back as a simulation-only extra.

External references (as cited in the NEXORA dossier; not independently re-verified here)

1. Smart India Hackathon, official problem statements workbook (SIH26-26007).
2. ISO 21815-1:2022, Earth-moving machinery, collision warning and avoidance, general requirements.
3. ISO 19014-1:2018, Earth-moving machinery, functional safety methodology.
4. ISO 19014-2:2022, Safety-related hardware and architecture requirements.
5. EMESRT, Vehicle Interaction Control Improvement.
6. UK Met Office, How visibility and meteorological optical range are measured.
7. Sheeny et al., RADIATE: A Radar Dataset for Automotive Perception in Bad Weather, ICRA 2021.
8. Sezgin et al., Safe Autonomous Driving in Adverse Weather: Sensor Evaluation and Performance Monitoring, IV 2023.
9. Indian Railways, Basic Concept of Signal Engineering (block and one-train-only principles).
10. ISA, ISA/IEC 62443 industrial automation and control system security standards.

All numbers in this document are illustrative examples or planning estimates until measured. ProofRoad is advisory decision support, not a certified safety system. Claims should be read as targets or hypotheses until they pass testing.